1 Core model and fast–slow structure

Consider a singularly perturbed system for turbulent kinetic energy E and mean shear S, with timescale separation $0 < \epsilon \ll 1$:

$$\epsilon \frac{dE}{dt} = f(E,S) = l\sqrt{E+\delta}\left(S^2 - \phi N^2 - c_b N^2 \frac{E}{E+\alpha}\right) - \frac{E^{3/2}}{l}, \tag{1}$$

$$\frac{dS}{dt} = g(E,S) = G - c_1 ES - c_2 S, \tag{2}$$

on the physical domain $\mathcal{D} = \{(E,S) \in \mathbb{R}^2 : E \geq 0\}$. Here l is the mixing length, N^2 the (constant) Brunt–Väisälä frequency, G the geostrophic forcing, $\phi > 0$ a stability-correction coefficient, $\delta, \alpha > 0$ regularization/saturation constants, $c_b = 0.50$, and $c_1 = 1.80\ \mathrm{s\,m^{-2}}$.

Show that the fast eigenvalue

$$\lambda_f(E,S) \equiv \frac{\partial f}{\partial E}(E,S) \tag{3}$$

determines the local stability of the critical manifold $\mathcal{C}_0 = \{(E,S) \in \mathcal{D} : f(E,S) = 0\}$, and classify branches of $\mathcal{C}_0$ into attracting ($\lambda_f < 0$), repelling ($\lambda_f > 0$), and fold ($\lambda_f = 0$) segments.

Derive an explicit closed-form expression for

$$\lambda_f(E,S) \equiv \frac{\partial f}{\partial E}(E,S), \tag{4}$$

showing all steps (product rule on $\sqrt{E+\delta}$, quotient/chain rule on $E/(E+\alpha)$, and the $E^{3/2}/l$ term).

You should obtain:

$$\lambda_f(E,S) = \frac{l\left(S^2 - \phi N^2\right)}{2\sqrt{E+\delta}} - c_b N^2 l\left[\frac{E}{2\sqrt{E+\delta}(E+\alpha)} + \frac{\alpha\sqrt{E+\delta}}{(E+\alpha)^2}\right] - \frac{3\sqrt{E}}{2l}. \tag{5}$$